

Simranjeet Kaur

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EDUCATION

Chandigarh University

Bachelor of Engineering in Computer Science Hons. (AI & ML)

Sep. 2022 – Present

CGPA: 8.69

Sant Karam Singh Academy

Higher Secondary Education(CBSE)

2020 – 2022

Percentage: 94%

PROJECTS

GrowthEngine: AI-Powered Software Engineering Simulation Platform 🔄

Feb 2026 – Present

Simulates real-world engineering by converting GitHub issues into structured coding workflows

- Architected a **full-stack monorepo system** (Next.js, Node.js, Supabase) to simulate real developer workflows end-to-end
- Built an automated pipeline to **ingest and transform real issues from GitHub** into structured coding problems with classification
- Developed a **hybrid evaluation engine** combining code execution (Judge0) with AI-based feedback for correctness, bugs, and optimization
- Engineered a **guided contribution system** enabling users to transition from problem-solving to real open-source PR workflows

SkyGuard: Cloudburst Risk Estimation System 🔄

Jan 2026

Spatiotemporal ML system for early detection of cloudburst events using satellite data

- Processed **20+ years of ERA5 and IMERG data** to engineer spatiotemporal features for extreme weather prediction
- Trained ensemble models (**Random Forest, XGBoost**) achieving **73% recall**, detecting 14/19 historical events
- Performed **lead-time analysis (6-24 hrs)** to optimize early-warning windows for practical deployment

ClinIQ-AI: Medical Report Intelligence Platform 🔄

July 2025

AI system for analyzing medical reports using OCR + RAG pipelines

- Built a **RAG-based pipeline** using FAISS and transformer embeddings for contextual medical insights
- Developed **multilingual conversational AI** to simplify complex medical reports for end users
- Designed **secure data pipelines** for report ingestion, storage, and personalized knowledge retrieval

ACHIEVEMENTS & CERTIFICATIONS

- Solved **200+ DSA problems** on LeetCode, strengthening algorithmic problem-solving **Link**
- Microsoft Azure AI Fundamentals Microsoft (2024) **Link**
- Finalist Bhasha Bandhu Hackathon (IEEE)

TECHNICAL SKILLS

Languages : Python, Java, SQL

Backend & Systems : Node.js, FastAPI, REST APIs, System Design

AI/ML : TensorFlow, Scikit-learn, RAG, LangChain

Databases : MySQL, MongoDB, Vector DBs, Supabase

Tools : Git, GitHub, Docker, Postman

Core : DSA, DBMS, OOP, CN, OS